Communication with an Adaptive Receiver Model: an Overview of a Conceptual Framework

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Abstract

Classical Shannon information theory treats the receiver as a passive decoder that reconstructs the transmitted message with minimum distortion. A real receiver, however, is not optimal: it has limited computational power, its own world model, a decision-making goal, and an individual history of interactions. This article proposes a conceptual framework that explicitly incorporates these aspects. Its main components are: (i) an extension of the rate-distortion function by the receiver's model and goal, (ii) learning the receiver's parameters from shared communication history, (iii) a probabilistic decision model that expresses, via a measurable function, the *cost of adaptation* and compares it against the expected benefit, thereby allowing a rational decision about whether this cost should be accepted at all, and (iv) thermodynamic limits of such adaptation. The central thesis is that optimizing transmission is not free: every efficiency gain requires computation, memory, time or energy, and sometimes this cost exceeds the value of the information gained. The aim is to identify open problems and prepare the ground for formal results; the article does not prove new theorems, but pinpoints where they might arise.

1 Introduction

Shannon's rate-distortion theory [1, 2] establishes a fundamental limit of transmission efficiency: a source X is compressed into R bits and the decoder reconstructs $\hat{X}$ with error $d(X, \hat{X}) \leq D$. The curve $R(D)$ is the minimum rate at a given distortion, and Shannon proved that it is achievable (with suitably long code blocks) independently of the receiver.

Wyner and Ziv [3] extended the theory to the case in which the receiver has side information S with a known joint distribution $P(X, S)$. The rate drops to

$$R_{WZ}(D) \;=\; \min_{P(U|X)} [I(X;U) - I(S;U)]$$

with reconstruction $\hat{X} = \kappa(U, S)$.

Both formulations, however, assume that the receiver is an optimal decoder with perfect knowledge of the statistics. This does not hold in practice. Language, human–machine dialogue, network communication between heterogeneous endpoints, federated learning, all are systems where the receiver is an *agent* with its own model, goal and constraints. This article outlines a framework that promotes these elements to first-class variables.

2 Receiver Model

We characterise the receiver by the triple (M, G, κ):

- M: the *world model*, an estimated distribution over X, in general different from the true P_X.
- G: the *goal*, a task with action space $\mathcal{A}$ and loss function $\ell(a, X)$.

- $\kappa_M : \mathcal{I} \to \mathcal{A}$: the *decision function* parametrised by the model M, not arbitrary.

The proposed rate-distortion function:

$$R_{M,G}(L) \;=\; \min_{f:\mathcal{X}\to\mathcal{I}} \mathbb{E}[|f(X)|] \quad \text{s.t.} \quad \mathbb{E}[\ell(\kappa_M(f(X)), X)] \le L. \tag{1}$$

Differences with respect to R_{WZ}:

1. distortion is measured by the *task loss* ℓ rather than $d(X, \hat{X})$,
2. κ is *fixed by the model* M, not freely optimised,
3. the expectation is taken under the true P_X, but κ uses the *perceived* model M, so model mismatch becomes an intrinsic component of the loss.

Informal decomposition of the expected loss:

$$\mathbb{E}[\ell] \;\approx\; \underbrace{L^\star(R)}_{\text{rate-limited}} \;+\; \underbrace{\Delta(M, P_X)}_{\text{model mismatch}} \;+\; \underbrace{\Delta(\kappa_M, \kappa^\star)}_{\text{bounded compute}} \;.$$

The second and third terms cannot be removed by increasing R, which is fundamentally new compared with Shannon.

3 History as the Source of Adaptation

The parameters (M, G, κ) are in general unknown and differ in practice between receivers. The only realistically available source of information about a particular receiver is the *history* of shared communication. By analogy with universal source coding [4, 5]:

$$R_T \;=\; R^\star \;+\; \frac{c(k)}{T}$$

where $R^\star$ is the optimal rate under full receiver knowledge, T is the length of history, and k the model complexity. For large T:

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} R_t \;=\; h(X, A), \tag{2}$$

where $h(X, A)$ is the *entropy rate* of the joint source and receiver-reaction process. This is an irreducible lower bound.

Communication therefore naturally splits into three phases:

1. *Cold phase*: universal code, history collection.
2. *Warm phase*: gradual adaptation to the estimated (M, G, κ).
3. *Steady state*: $R_T \to R_{M,G}(L)$, the penalty $c(k)/T$ vanishes.

In the case of bidirectional interaction [6, 7] convergence is faster, because every exchange carries two signals (content plus an indirect signal about the goal).

If we have access to the history of *many* receivers, we can estimate a *population manifold* in the parameter space. For a new receiver it then suffices to localise it with a few observations, and the adaptation penalty drops to $c(d)/T$, where $d \ll k$ is the intrinsic dimension of the manifold (hierarchical Bayes, meta-learning).

4 Decision Framework: When to Adapt

The central thesis of this article is that optimizing transmission is not free. Adaptation costs time $\tau(T)$, memory, energy and computation. We therefore need a *measurable function* that quantifies the cost of adaptation and places it against the expected benefit, so that we can rationally decide whether to accept this cost at all. Without such a function, optimization becomes a dogma, even though in some cases its cost exceeds the value of the information gained.

We define the expected benefit and the expected cost:

$$\text{Benefit} = N_{\text{future}} \cdot [R_{\text{univ}} - R_{\text{adapt}}(T)] \cdot c_{\text{bit}},$$

$$\text{Cost}(T) = C_{\text{comp}} \cdot \tau(T) + C_{\text{energy}} + C_{\text{mem}}.$$

Deterministic decision rule:

$$\text{adapt} \iff \text{Benefit} > \text{Cost}.$$

Since all input quantities (gap, relationship length, τ) are uncertain and no two receivers are identical, a probabilistic version is more natural. Define:

$$P_{\text{improve}}(k) \;=\; \Pr\big[\, G \cdot N \cdot c_{\text{bit}} > C_{\text{comp}} \cdot \tau \,\big|\, \text{obs}_{1:k} \,\big], \tag{3}$$

where G is the gap $H(X) - h(X, A)$ and N the expected number of future messages. The function $P_{\text{improve}}(k) \in [0, 1]$ is the *measurable output* that directly underpins the decision:

$$\text{adapt} \iff P_{\text{improve}}(k) > \theta.$$

The threshold θ expresses risk tolerance and willingness to pay an uncertain cost.

Under a Gaussian prior on the population (G, N), the function P_{improve} admits a closed form via the cumulative distribution function Φ, so it is *exactly computable* from statistics of the history and the population. The accuracy of population parameter estimates grows as $1/\sqrt{N_{\text{pop}}}$.

The function $P_{\text{improve}}(k)$ cannot answer for $k = 0$ without a population prior, since the decision model always requires either individual history or amortised population experience. Computing P_{improve} also has a cost, which must be lower than the average saving from the decisions it produces; otherwise the decision model becomes yet another problem that needs its own decision (the principle of *selective adaptation*).

5 Thermodynamic Limits

Every model adaptation involves memory operations. Landauer's principle [8] sets a lower bound on the energy of an irreversible bit change:

$$E_{\min} = kT \ln 2.$$

In practice modern systems operate about $10^{10}\times$ above the Landauer limit, so the thermodynamic component is negligible. In the limits (reversible and quantum systems, high-density memory) it becomes dominant.

Globally, the second law of thermodynamics implies that every local reduction of information entropy is paid for by a larger rise of entropy in the environment. Information efficiency is not free even in the most basic physical sense, because it is always paid for by heat dissipated into the surroundings.

6 Boundaries of the Model

The model is an abstraction over the symbolic and semantic layers of communication; the physical layer (analogue quantities, modulation, channel noise) is assumed to be solved through the classical *channel coding theorem*. By "data" we mean the bits transmitted through the channel; information, knowledge and wisdom arise only at the receiver through the interaction of data with its model.

Full efficiency of transmission is unreachable for five independent reasons: (1) channel noise, (2) incomplete knowledge of the receiver, (3) finite transmission time, (4) finite adaptation time, (5) the irreducible entropy $h(X, A)$ of the joint process. The framework focuses on points (2) and (4).

7 Open Questions

- **Gap theorem**: does a tight upper/lower bound on $R_{M,G}(L) - R_{WZ}(L)$ exist as a function of $\mathrm{KL}(P_X \| M)$?
- **Rate-distortion for the receiver manifold**: what is the optimal rate for representing a receiver with a given quantisation error? This question connects Shannon's rate-distortion theory with manifold learning and nonparametric Bayesian methods.
- **Optimal stopping**: is there a closed form for the threshold $\theta^\star(\tau, N)$ in the probabilistic decision (3)?
- **Task-aware entropy rate**: for which class of goals G does the entropy rate $h(X, A)$ drop below $h(X)$? Can these classes be characterised in information-geometric terms?
- **Equivalence with the information bottleneck** [9]: is the categorisation of receivers equivalent to an IB problem with task loss as relevance?

8 Conclusion

The presented framework consolidates several strands of thought: rate-distortion with side information, universal source coding, semantic communication [10], meta-learning and online adaptation. None of the individual results is new, what is new is their *integrated framing* with an explicit decision module and a thermodynamic perspective. The central motif is the construction of a measurable function for the cost of optimization: without it, adaptation is done blindly; with it, it becomes a rational decision that sometimes reads "do not adapt", because nothing in the world is free. The aim of further work is to convert the open questions into formal claims and to prove at least one non-trivial theorem.

References

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